

Constants

$$c = 299\,792\,458 \text{ m/s} = \lambda f \quad k = 1.381 \times 10^{-23} \text{ J/K}$$

$$R_{\oplus} = 6378.137 \text{ km} \quad \mu_{\oplus} = 398\,600.5 \text{ km}^3/\text{s}^2 \quad M_{\oplus} = 7.96 \times 10^{22} \text{ kg}$$

$$273.15 \text{ K} = 0^\circ \text{C} \quad \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

$$1 \text{ AU} = 149\,597\,870.7 \text{ km} \quad S_{\odot} = \frac{L_{\odot}}{4\pi R^2} = 1358 \text{ W/m}^2 \text{ at } 1 \text{ AU}$$

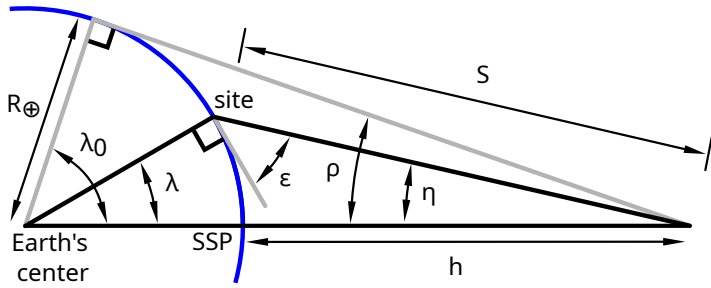
$$L_{\odot} = 3.83 \times 10^{26} \text{ W (total solar output)}$$

Orbits

$$\varepsilon = \frac{V^2}{2} - \frac{\mu}{R} = \frac{-\mu}{2a} \quad R = \frac{a(1-e^2)}{1-e\cos\nu} \quad V_{\text{circ}} = \sqrt{\frac{\mu}{R}}$$

$$\mathbb{P} = 2\pi\sqrt{\frac{a^3}{\mu}} \quad n = \sqrt{\frac{\mu}{a^3}} = \frac{2\pi}{\mathbb{P}} \quad R = R_{\oplus} + h$$

Mission Geometry

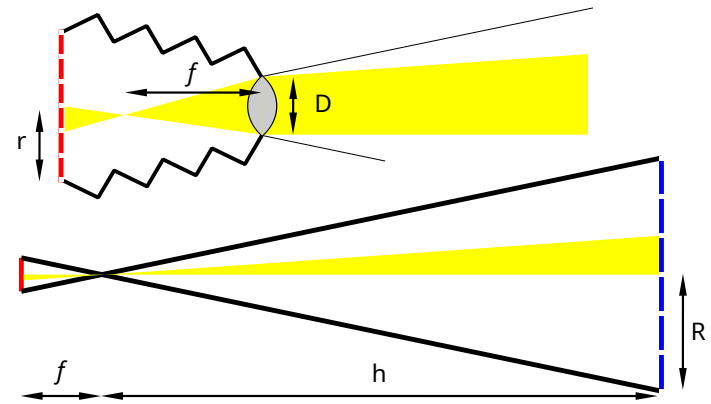


$$\sin \rho = \cos \lambda_0 = \frac{R_{\oplus}}{R_{\oplus} + h} \quad \lambda_0 + \rho = 90^\circ \quad \eta + \lambda + \varepsilon = 90^\circ$$

$$\tan \eta = \frac{\sin \rho \sin \lambda}{1 - \sin \rho \cos \lambda} \quad \sin \eta = \cos \varepsilon \sin \rho$$

$$S = R_{\oplus} \frac{\sin \lambda}{\sin \eta} = -R_{\oplus} \sin \varepsilon + \sqrt{(R_{\oplus} \sin \varepsilon)^2 + h^2 + 2R_{\oplus}h}$$

Remote Sensing



$$Res = \frac{2.44\lambda h}{D} \quad Res = \frac{2R}{\# \text{ pixels}} \quad mag = \frac{f}{h} = \frac{r}{R}$$

$$\lambda_{max} = \frac{2.898 \times 10^{-3}}{T}$$

Electrical Power

$$T_e = \mathbb{P} \frac{2\rho}{360^\circ} \quad P_{req} = \frac{P_e T_e / \eta_e + P_d T_d / \eta_d}{T_d} \quad C_{batt} = \frac{P_e T_e}{DOD \eta_{batt}}$$

$$P_{BOL} = S_{\odot} A \eta \cos i \quad P_{sa} = P_{BOL} (1 - \text{degradation})^{time}$$

Communication

$$X \text{ (dB)} = 10 \log_{10} x \text{ (linear)} \quad 10^{X \text{ (dB)/10}} = x \text{ (linear)}$$

$$\frac{E_b}{N_0} = \frac{P_t G_t L_l L_s L_m G_r}{k T_r R} = SNR \frac{B}{R} \quad EIRP = P_t L_t G_t$$

$$\frac{E_b}{N_0} = P_t + G_t + L_l + L_s + L_m + G_r + 228.6 - 10 \log_{10} T_r - 10 \log_{10} R$$

$$L_s = \left(\frac{\lambda}{4\pi S} \right)^2 \quad L_m = L_{pt, tx} + L_{pt, rx} + L_{pol} + L_{atm} + L_{rain}$$

$$L_{pt} \text{ (dB)} = -12 \left(\frac{\theta}{\theta_{3dB}} \right)^2 \quad L_{pol} = \cos^2 \theta \text{ (linear pol only)}$$

$$L_{atm} \text{ (dB)} \approx \frac{A_{zenith} \text{ (dB)}}{\sin \varepsilon} \quad \theta_{3dB, par} \text{ (deg)} = 70 \frac{\lambda}{D_{par}}$$

$$G_{par} = \left(\frac{\pi D_{par}}{\lambda} \right)^2 \eta \quad \eta_{par} = 0.55-0.7 \text{ (typical)}$$

Attitude

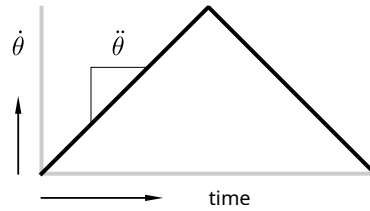
$$\vec{T} = I \vec{\alpha} \quad \int \vec{T} dt = \Delta \vec{H} \quad \sum T_{ext} = T_{SRP} + T_{drag} + T_m + T_g$$

$$T_{SRP} = F_{SRP} (c_{ps} - c_g) \quad F_{SRP} = \frac{S_{\oplus}}{c} A_s (1 + q) \cos i$$

$$T_{drag} = F_{drag} (c_{pa} - c_g) \quad F_{drag} = \frac{1}{2} \rho C_d A V^2$$

$$\vec{T}_m = \vec{D} \times \vec{B} \quad D = NIA \quad B_{pole} \approx \frac{2M}{R^3} \quad B_{equator} \approx \frac{M}{R^3}$$

$$T_g = \frac{3\mu}{2R^3} |I_{yaw} - I_{other}| \cos 2\theta$$



Reaction Wheels

$$\vec{H} = I \vec{\omega} \quad I_{cylinder, z} = \frac{1}{2} m r^2 \quad T_{slew} = \frac{4I\theta}{t^2}$$

Thrusters

$$F_t = \frac{T_D}{L} \quad F_{dump} = \frac{h}{L t}$$

Thermal

$$Q_{out} + P_{out} = Q_{in} + Q_{internal} \quad Q = \varepsilon \sigma A T^4 \quad Q_{\odot} = S_{\odot} A \alpha \cos i$$

$$Q_{albedo} = S_{albedo} A \alpha \cos i \left(\frac{R_{\oplus}}{R_{\oplus} + h} \right)^2 \quad S_{albedo} = 407 \text{ W/m}^2$$

$$Q_{\oplus IR} = S_{\oplus} A \varepsilon \cos i \left(\frac{R_{\oplus}}{R_{\oplus} + h} \right)^2 \quad S_{\oplus IR} = 237 \text{ W/m}^2$$

Reliability

$$R = e^{-\lambda t} \quad \lambda = \frac{1}{MTBF} \quad F = 1 - R$$

$$\text{series : } R = R_A R_B \quad \text{parallel : } R = 1 - [(1 - R_A)(1 - R_B)]$$